

Homework 7, 2026

This homework needs to be done and submitted in groups of 3 to 5 students.

Please note: in this assignment statement the notation of MLS is used a few times (not necessarily always) without explanation where it is used.

Consider a four degree of freedom robotic manipulator schematically shown in figure 1, in its reference configuration. This is exactly the manipulator discussed in MLS example 4.4 p. 177.

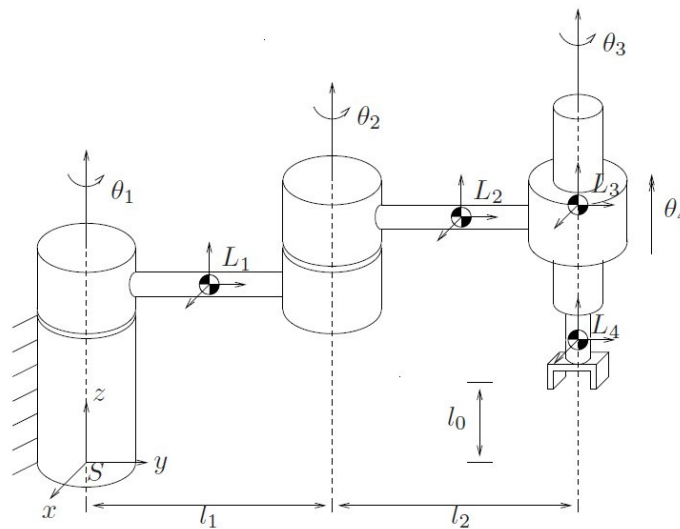


Figure 1: The manipulator in the reference configuration.

Assignment:

- (a) Build a SIMULINK model of a computed torque controller controlling the manipulator for tracking a desired joint angle trajectory θ_d , which is in the form of a rounded step function. To do this, you should execute all steps specified in the May 27 and June 3, 2026, tutorials. (Note that you are allowed to do this whole homework directly in either MATLAB or Python, but it is easier to do it in SIMULINK.)

The three revolute joints, joints 1, 2 and 3, should track a final values of 2, 1 and 0.5 radians, respectively, while the prismatic joint should track a final value of 0.08 m. Since for the computed torque control law θ_d must be twice differentiable, a scalar unit rounded step function that meets this requirement, is given to you on Moodle under the June 3 tutorial. This function generates the rounded step as well as its 1st and 2nd time derivatives. You should be able to use it as is, in your SIMULINK model, except for the need to adjust the step sizes. To get the rounded step into your model, simply select it inside a open SIMULINK window, copy it to the paste buffer and then paste it anywhere in the open SIMULINK window of your model. If you decide to do the homework in MATLAB or Python, you can contact me to get copies of the MATLAB or Python code for the rounded step calculation.

You need to use the numeric parameters given to you on Moodle for the May 27, 2026 tutorial, except for the eight values θ_i and $\dot{\theta}_i$, $i = 1, 2 \dots 8$, which now will be feedback variables of time.

- (b) Once you are confident that your controller works and the simulation runs well, use your simulation to study the tracking robustness of the computed torque control scheme. To do this, you may for instance reduce or increase all the values in the $\mathcal{M}'$ matrix by 5 or 10 percent **in the plant only**, while keeping the corresponding values in the controller at their nominal values. This corresponds to the scenario where the plant parameters have changed due to aging, while the controller parameters have not changed. Report on this investigation.

Note: to do this correctly, you need to have two function blocks to calculate the $M(\theta)$, $C(\theta, \dot{\theta})$ and $N(\theta, \dot{\theta})$ matrices in your SIMULINK program:

- one using the original robot parameters, representing the controller that has not changed with aging, and passing its output data to the computed torque calculation, and
 - one using the slightly changed robot parameters, representing the robot arm that has changed with aging, and passing its output data to the ODE solver that needs to solve the differential equations of motion in simulation.
- (c) Also investigate the use of a PD control scheme on this manipulator and report on your findings.
- (d) Returning to the computed torque control scheme, with the plant and controller having the same properties: it is observed that the torque values that are required to track the required θ_d trajectory are quite large. Investigate the effect of torque saturation. For this assume that the motor torques are limited to the following values: $|\tau| \leq [6000 \ 2000 \ 1000 \ 100]^T$ Nm or N.

Include graphs of simulated results in your report, especially to support your arguments or as supporting evidence of observed phenomenon.

You need to upload your report to the Moodle assignment.